Hsuan-Fu (Jeff) Hsu

Embedded Systems Engineering | C/C++, Sensor Integration & Firmware, Hardware-Software Systems

Los Angeles, CA | 602-520-6877 | hsuanfu0709@gmail.com | LinkedIn | GitHub | Portfolio

EDUCATION

University of Southern California — Los Angeles, CA
Master of Science in Computer Science | GPA: 3.6/4.0 — Jan 2026 - May 2027 (Expected)

National Taipei University — New Taipei City, Taiwan
Bachelor of Science in Computer Science | GPA: 3.5/4.0 — Sept 2020 - June 2024

SKILLS

Languages: C++, C, C#, Python, SQL

Embedded & Hardware: Arduino (Nano 33 BLE Sense), I2C sensor configuration (AS7263), BLE wireless communication, analog/PPG signal acquisition, DC-DC power regulation, 3D-printed enclosure design

Systems & Graphics: OpenGL, GLSL, manual GPU buffer/memory management, real-time rendering pipelines

Software Engineering: Git, GitHub Actions CI, xUnit, pytest, ASP.NET Core, SQL Server

EXPERIENCE

National Taipei University (NTPU) — New Taipei City, Taiwan
Undergraduate Researcher | Advisor: Prof. Bor-Shing Lin — July 2022 - Feb 2024

- **Co-designed and built a wearable dual-wavelength NIR biosensing device** with 2 other core contributors: an **Arduino Nano 33 BLE Sense** microcontroller, an **AS7263 NIR sensor** (810/860 nm) configured over I2C (gain, integration time), and BLE transmission to a companion mobile app.
- **Wrote embedded firmware** for PPG signal acquisition and serial/BLE data streaming; co-designed the 3D-printed device enclosure and the DC-DC boost converter power stage.
- Co-designed the experimental protocol and led model selection across 5 ML and 4 DL architectures using leave-one-subject-out cross-validation; best model achieved **10.3% MAPE**. Coauthor, IEEE Transactions on Human-Machine Systems (2026).

Tech-Management Workshop Co., Ltd. — Taichung, Taiwan
Software Engineer Intern — July 2026 - Aug 2026

- Independently designed and built a government birth grant system with **ASP.NET Core MVC** and **SQL Server**, now deployed across 29 household registration offices.
- Added server-side validation, CSRF protection, and HTTPS/HSTS; reduced database latency by **at least 50%** and overall resource usage by **at least 30%** compared with the previous implementation.

PROJECTS

3D Rendering Engine [GitHub] — Self-directed
C++, OpenGL, GLSL

- Built a real-time OpenGL/GLSL rendering engine from scratch in **C++**, including a custom OBJ/MTL parser, multi-light Phong shading, and skybox environment mapping with no external model-loading library.
- **Managed GPU buffers (VAO/VBO) and heap-allocated scene objects explicitly**, with deterministic resource cleanup on exit — no framework lifecycle management.

Task Management System [GitHub] — Jan 2026 - Sept 2026
ASP.NET Core, Vue 3, TypeScript

- Built 7 REST endpoints with **ASP.NET Core** and a Vue 3/TypeScript UI, with JWT authentication, ownership checks, and dashboard summaries.
- Wrote 6 **xUnit** integration tests; automated CI with GitHub Actions and packaged the UI and API with Docker.

PUBLICATION

Coauthor, "Noninvasive Portable System With Dual Near-Infrared Light Technology for Monitoring Uric Acid Concentration," IEEE Transactions on Human-Machine Systems, accepted for publication, Aug 2026.